

IoT-Enabled Retail Analytics on Azure Databricks

creation of storage account:

The screenshot displays the Azure portal interface for a storage account named 'sagarcatalog'. The left sidebar shows the 'Storage accounts' section with a list of accounts: 'databricksmentor', 'nilaystore', 'sagarcatalog' (selected), and 'sagarstorage0'. The main content area shows the 'Overview' tab for 'sagarcatalog'. The 'Essentials' section lists key properties: Resource group (LabsKraft2027), Location (westus2), Subscription (LabsKraft), Subscription ID (d008c7cc-9795-4292-bf79-74ae52cda63d), Disk state (Available), and Tags (Add tags). The 'Properties' section shows 'Data Lake Storage' settings: Hierarchical namespace (Enabled), Default access tier (Hot), and Blob anonymous access (Disabled). The 'Security' section shows 'Require secure transfer for REST API operations' (Enabled) and 'Storage account key access' (Enabled).

creation of access connector:

The screenshot displays the Azure portal interface for an access connector named 'sagarconnector'. The left sidebar shows the 'Access Connectors for Azure Databricks' section with a list of connectors: 'sagarconnector' (selected). The main content area shows the 'Overview' tab for 'sagarconnector'. The 'Essentials' section lists key properties: Resource group (LabsKraft2027), Location (West US 2), Subscription (LabsKraft), Subscription ID (d008c7cc-9795-4292-bf79-74ae52cda63d), and Tags (Add tags). The 'State' is 'Succeeded' and the 'Resource ID' is '/subscriptions/d008c7cc-9795-4292-bf79-74ae52cda63d/resourceGroups/...'. The 'Settings', 'Automation', and 'Help' sections are also visible.

giving access connector permission for storage:

The screenshot shows the Microsoft Azure portal interface for adding a role assignment. The main pane is titled "Add role assignment" and shows the "Storage Blob Data Contributor" role selected. The "Assign access to" section has "Managed identity" selected. The "Members" section shows "No members selected". The "Description" field is optional. The sidebar titled "Select managed identities" shows a search for "Access Connector for Azure Databricks (1)" and lists the identity "sagarconnector" with its object ID. The "Selected members" section shows "No members selected".

creation of uc:

The screenshot shows the Microsoft Azure Databricks interface. The left sidebar shows the "Catalog" section. The main pane shows the "Catalog Explorer" for "sagar_cat_project1". The "About this catalog" section shows details like Metastore Id, Created At, Created By, Updated At, Updated By, Storage Root, and Storage Location. The "Advanced" section shows "Predictive Optimization" is enabled.

creation of namespace and eventhub:

[Home](#) > [Event Hubs](#) >



Create Namespace ...

Event Hubs

Project Details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	LabsKraft
Resource group *	LabsKraft2027

[Create new](#)

Instance Details

Enter required settings for this namespace, including a price tier and configuring the number of units (capacity).

Namespace name *	sagarspace
Region *	West US 2
Pricing tier *	Basic
Throughput Units *	1

[Browse the available plans and their features](#)

[Review + create](#)

[< Previous](#)

[Next: Advanced >](#)

portal.azure.com/#view/Microsoft_Azure_EventHub/CreateBlade/_provisioningContext~/%7B%22initialValues%3A%7B%22subscriptionId%3A%5B%22d008c7cc-9795-4292-bf79-74ae52c...

Microsoft Azure

Search resources, services, and docs (G+/)

Copilot

tredecence14@michaelklo...
DEFAULT DIRECTORY (MICHAELKLO...)

Home > Event Hubs >

Create Namespace

Event Hubs

Validating...

BasicsAdvancedNetworkingTagsReview + create

Event Hubs Namespace
by Microsoft

Basics

Namespace name	sagarhub
Subscription	LabsKraft
Resource group	LabsKraft2027
Region	West US 2
Pricing tier	Basic
Throughput Units	1
Availability Zones (Zone Redundancy)	Enabled

Networking

Connectivity method	Public access
---------------------	---------------

Security

Minimum TLS version	1.2
---------------------	-----

Create< PreviousNext >

24°C
Mostly cloudy

Search

Copilot

tredecence14@michaelklo...
DEFAULT DIRECTORY (MICHAELKLO...)

Home > Event Hubs > sagarspace >

Create Event Hub

Event Hubs

Validation succeeded.

BasicsCaptureReview + create

Event Hubs Instance
by Microsoft

Basics

Name	sagarhub
Partition count	1

Retention

Cleanup policy	Delete
Retention time (hrs)	6

Capture

Capture Status	Not Supported
----------------	---------------

Create< PreviousNext >

creating synapse workspace:

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

🔍 🔔 ⚙️ 🔄 🗨️

FREQUENTLY USED
DEFAULT DIRECTORY

[Home](#) > [Azure Synapse Analytics](#) >

Create Synapse workspace

✔️ Validation succeeded

Information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#).

Basics

Subscription	LabsKraft
Resource group	LabsKraft2027
Region	West US 2
Workspace name	(new) sagars
Data Lake Storage Gen2 account	https://sagarstorage0.dfs.core.windows.net
Data Lake Storage Gen2 file system	raw
Managed resource group	None
Role assignments	The Storage Blob Data Contributor role will be assigned on the specified Data Lake Storage Gen2 account to both the workspace managed identity and the current user.

Security

Authentication method	Use both local and Microsoft Entra ID authentication
SQL Server admin login	sqladminuser
SQL Password	*****
Double encryption	No

Create

< Previous

Next >

Download a template for automation

setting up synapse ->

linking my adls for storing old data and these can be pulled into azure dbx.

Data

Workspace Linked

Filter resources by name

- Azure Data Lake Storage Gen2 2
 - sagars (Primary - sagarstorage0)
 - raw (Primary)**
 - (Attached Containers)

raw

Other users in your workspace may have access to modify this item. Do not use this item unless you trust all users who may have access to the workspace.

New SQL script New notebook New data flow New integration dataset Upload Download New folder More

← → ↓ ↑ raw

Name	Last Modified	Content Type	Size
retail_jot_data.csv	8/25/2025, 10:26:07 AM		130.8 KB

creation of new dedicated pool:

Microsoft Azure | Synapse Analytics | sagars

Synapse live | Validate all | Publish all

New dedicated SQL pool

Validation succeeded.

Basics * Additional settings * Tags **Review + create**

Product details

Azure Synapse Analytics dedicated SQL pool by Microsoft
[Terms of use](#) | [Privacy policy](#)

Est. cost per hour: 99.83 INR
[View pricing details](#)

Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#)

Data source

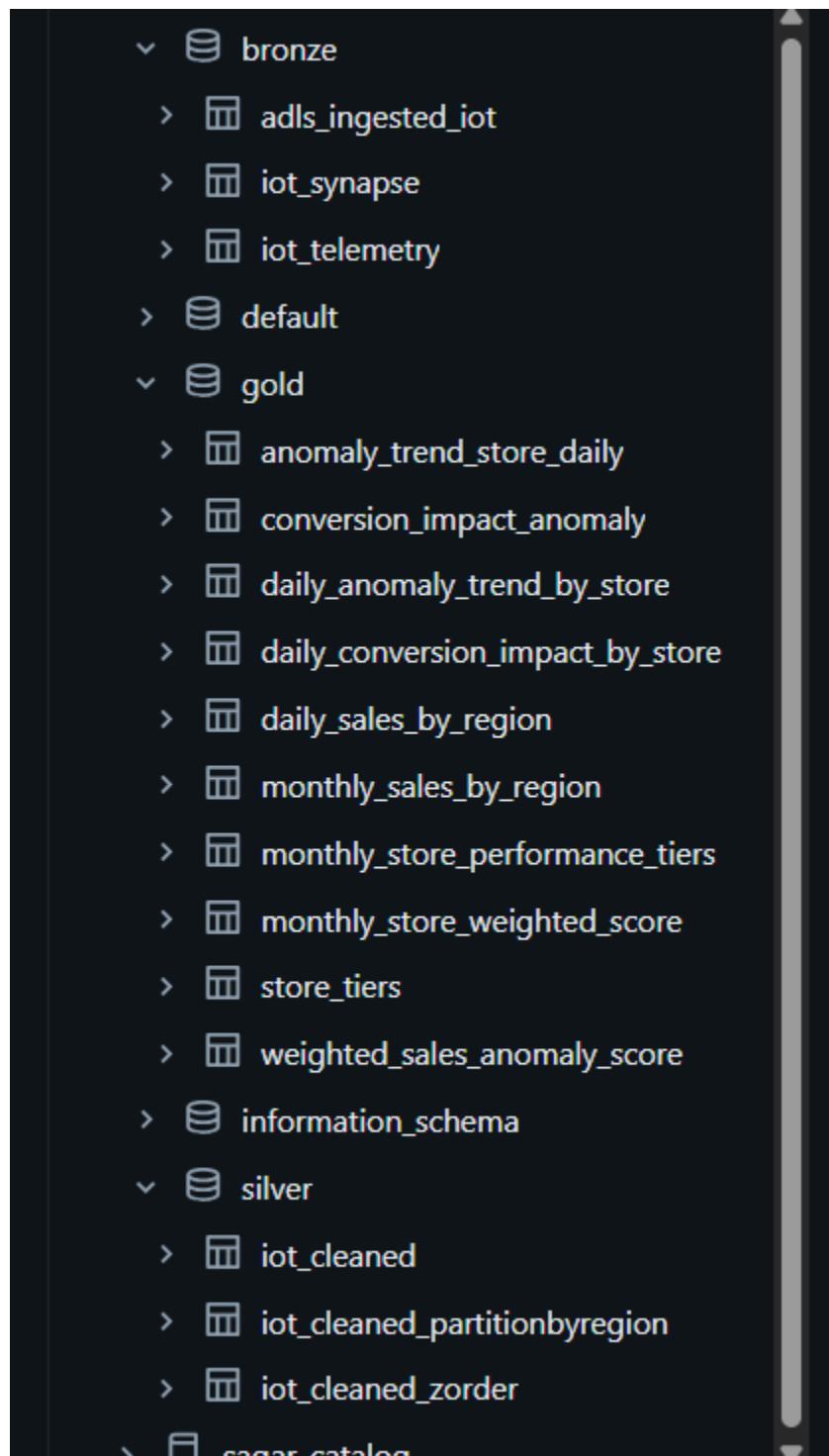
Dedicated SQL pool name	db1
Performance level	DW100c

Additional settings

Use existing data	Blank
Collation	SQL_Latin1_General_CP1_CLAS

Create < Previous Download template for automation Cancel

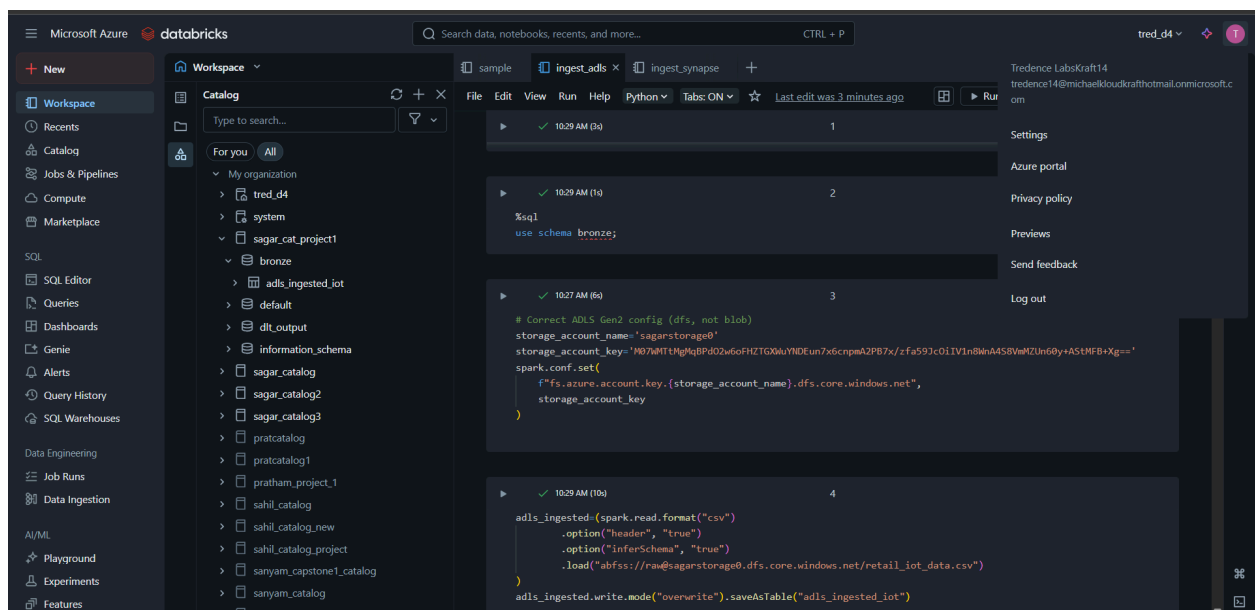
Quick Summary->



Ingestion mode->

i) ADLS

adls connection and storing file in sagar_cat_project1.bronze schema



adls_ingested_table->

```
option("inferSchema", "true")
.load("abfss://raw@sagarstorage0.dfs.core.windows.net/retail_iot_data.csv")

adls_ingested.write.mode("overwrite").saveAsTable("adls_ingested_iot")
display(adls_ingested)
```

(7) Spark Jobs

adls_ingested: pyspark.sql.connect.dataframe.DataFrame = [OrderID: string, CustomerID: string ... 12 more fields]

OrderID	CustomerID	Region	StoreID	Product	Quantity	UnitPrice	SalesAmount	OrderTime	Device
db07ecdd-e8e7-408a-846e-c2d330484d4b	CUST-2824	East	Store-009	Mobile	2	322.1	644.2	2023-01-05T23:24:55.000+00...	Temp_Sc
93d89258-6e02-4bb8-9a71-cb590cb55f85	CUST-7912	East	Store-001	Laptop	2	503.69	1007.38	2023-01-30T05:24:09.000+00...	RFID_Sc
f97636e9-3dab-48ad-aad4-0ca034d3111	CUST-9928	South	Store-008	Tablet	5	592.47	2962.35	2023-01-01T07:34:17.000+00...	RFID_Sc
i20cae76-3e09-4580-8429-af1d6b76209a	CUST-3547	West	Store-011	Laptop	1	790.86	790.86	2023-01-18T10:14:28.000+00...	POS_Ter
5dbc9ddc-0275-4427-a4b6-089b43465200	CUST-8527	East	Store-013	Laptop	11	621.7	6838.7	2023-01-18T13:20:07.000+00...	Temp_Sc
c6bd396c-d239-42b7-b414-4e39289bc2da	CUST-5741	East	Store-008	Printer	1	791.25	791.25	2023-01-23T00:15:40.000+00...	Temp_Sc
02fadf02-d15e-4527-94b1-9cb915a71209	CUST-6820	West	Store-009	Smartwatch	1	1237.81	1237.81	2023-01-09T07:23:27.000+00...	Temp_Sc
90e7e190-9d09-47a3-a399-98562cf389dc	CUST-7216	North	Store-018	Mobile	3	1693.56	5080.68	2023-01-03T17:10:20.000+00...	RFID_Sc
50f6c840-47fc-4117-a776-78701a911199	CUST-7572	North	Store-003	Mobile	5	1758.92	8794.6	2023-01-16T06:37:34.000+00...	RFID_Sc
77d4cd70-6019-4be0-bce4-51626ae52101	CUST-8517	West	Store-009	Mobile	2	1502.73	3005.46	2023-01-27T03:57:14.000+00...	POS_Ter
6d65d67c-bc5e-43bf-9522-50e645af5f8f	CUST-7543	North	Store-008	Mobile	5	1012.36	5061.8	2023-01-03T06:53:40.000+00...	RFID_Sc
d6c5b3f1-076e-4c24-a836-318d058beedd	CUST-7916	East	Store-013	Tablet	5	1992.44	9962.2	2023-01-26T16:27:46.000+00...	POS_Ter
18175f8f-6017-413a-9e20-c876857ee103	CUST-1188	East	Store-018	Printer	3	1548.77	4646.31	2023-01-17T12:19:59.000+00...	RFID_Sc
54a45330-e78f-4d3d-9426-8eb38292bbd3	CUST-1053	North	Store-017	Printer	2	1039.98	2079.96	2023-01-06T03:58:39.000+00...	Temp_Sc

ii)Event Hub:

we will connect to vm and then send data from local machine to eventhub which at the end will be recieved by DBX

creation of vm (namespace and eventhub already created) ->

Microsoft Azure

Home > Compute infrastructure | Virtual machines >

Create a virtual machine

Validation passed

Preferred e-mail address: tredence14@michaelklo...

Preferred phone number:

You have set SSH port(s) open to the internet. This is only recommended for testing. If you want to change this setting, go back to Basics tab.

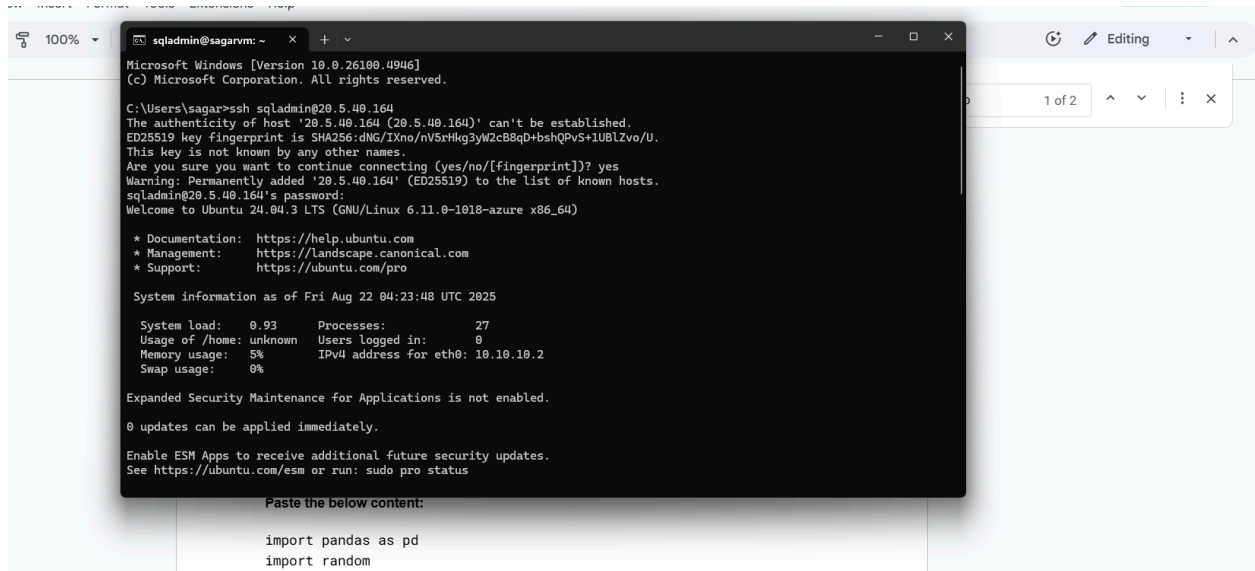
Basics

Subscription	LabsKraft
Resource group	LabsKraft2027
Virtual machine name	sagarvm
Region	Australia East
Availability options	Availability zone
Zone options	Self-selected zone
Availability zone	1
Security type	Trusted launch virtual machines
Enable secure boot	Yes
Enable vTPM	Yes
Integrity monitoring	No

< Previous Next > Create

Download a template for automation Give feedback

setting up vm session on local machine->



```
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sagar>ssh sqladmin@20.5.40.164
The authenticity of host '20.5.40.164 (20.5.40.164)' can't be established.
ED25519 key fingerprint is SHA256:dNG/IXho/nV5rHkg3yW2cB8qD+bshQpVs+1UblZvo/U.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '20.5.40.164' (ED25519) to the list of known hosts.
sqladmin@20.5.40.164's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.11.0-1018-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Aug 22 04:23:48 UTC 2025

System load:  0.93      Processes:    27
Usage of /home: unknown  Users logged in:  0
Memory usage: 5%       IPv4 address for eth0: 10.10.10.2
Swap usage:  0%

Expanded Security Maintenance for Applications is not enabled.

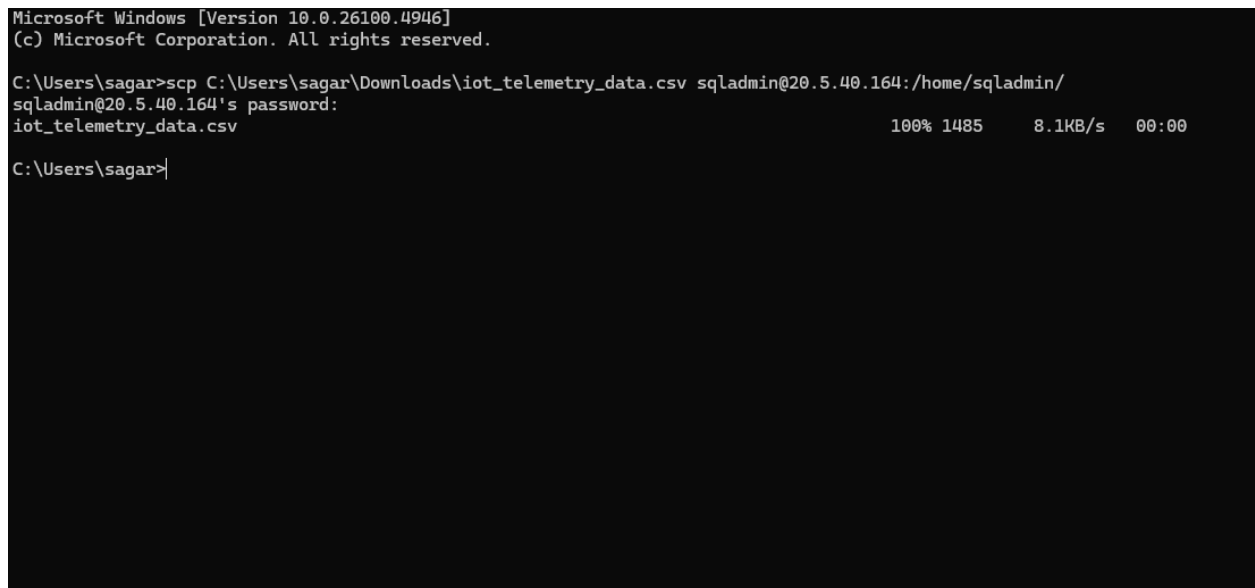
0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Paste the below content:

import pandas as pd
import random
```

uploading my file to vm which is considered for ingestion



```
Microsoft Windows [Version 10.0.26100.4946]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sagar>scp C:\Users\sagar\Downloads\iot_telemetry_data.csv sqladmin@20.5.40.164:/home/sqladmin/
sqladmin@20.5.40.164's password:
iot_telemetry_data.csv                                100% 1485      8.1KB/s   00:00

C:\Users\sagar>
```

nano [send.py](#)

```
GNU nano 7.2 send.py
ls /home/sqladmin/ from azure.eventhub import EventHubProducerClient, EventData
import pandas as pd
import time
import json

# Load the CSV
df = pd.read_csv('hot_telemetry_data.csv')

# Event Hub connection
connection_str = "Endpoint=sb://sagarspace.servicebus.windows.net/;SharedAccessKeyName=RootManageSharedAccessKey;SharedAccessKey=2v07F8+tBbuM02M5qjzsSpUAVWbYgfeZ+AEhBrynDE="
eventhub_name = "sagarhub"

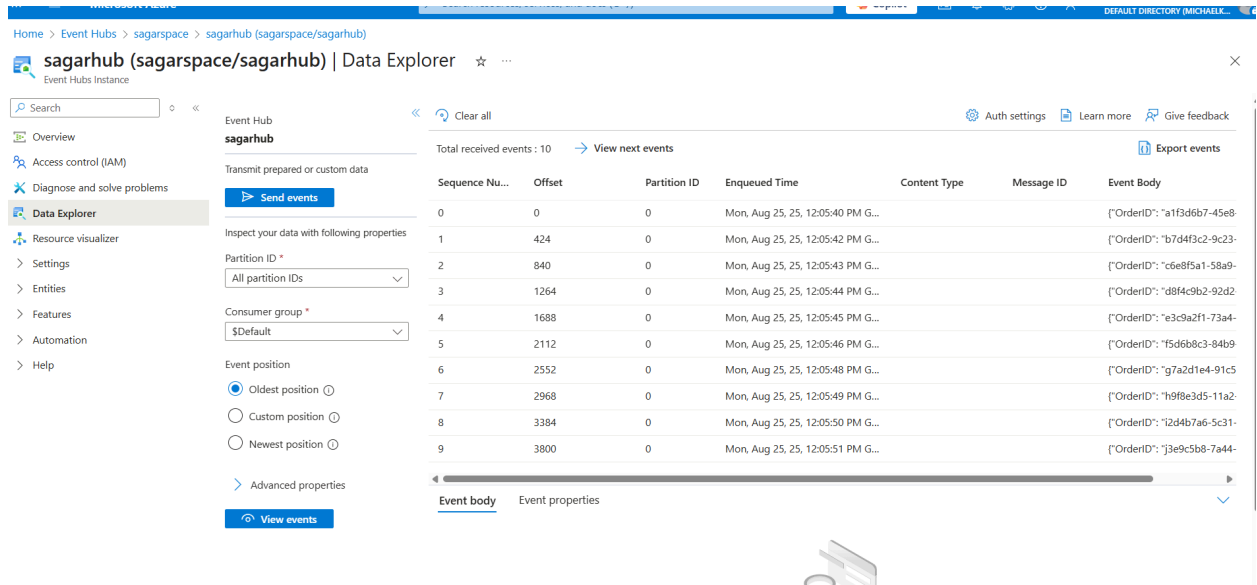
producer = EventHubProducerClient.from_connection_string(conn_str=connection_str, eventhub_name=eventhub_name)

# Stream data row by row
for _, row in df.iterrows():
    data = row.to_dict()
    event_data_batch = producer.create_batch()
    event_data_batch.add(EventData(json.dumps(data)))
    producer.send_batch(event_data_batch)
    print(f'Sent: {data["OrderID"]}')
    time.sleep(1) # simulate 1 event/sec
```

sent file:

```
WARNING: The script faker is installed in '/home/sqladmin/.local/bin' which is not
Consider adding this directory to PATH or, if you prefer to suppress this warning,
Successfully installed faker-37.5.3 numpy-2.3.2 pandas-2.3.2 tzdata-2025.2
sqladmin@sagarvm:~$ python3 send.py
Sent: a1f3d6b7-45e8-4a30-8e9b-1f9d2caa6710
Sent: b7d4f3c2-9c23-4a0a-a4b5-f0317b53f8f5
Sent: c6e8f5a1-58a9-4980-a71c-34266c5db412
Sent: d8f4c9b2-92d2-4a9f-a6b0-15db9d214e62
Sent: e3c9a2f1-73a4-41e8-b4fd-62b0a930ed7b
Sent: f5d6b8c3-84b9-4a77-a9c1-9d1e1cbf6d2e
Sent: g7a2d1e4-91c5-4f21-bf28-7bc0e55d9183
Sent: h9f8e3d5-11a2-48a4-9c2b-64b0b11c6d45
Sent: i2d4b7a6-5c31-4c29-bd38-27f7a04d0c9e
```

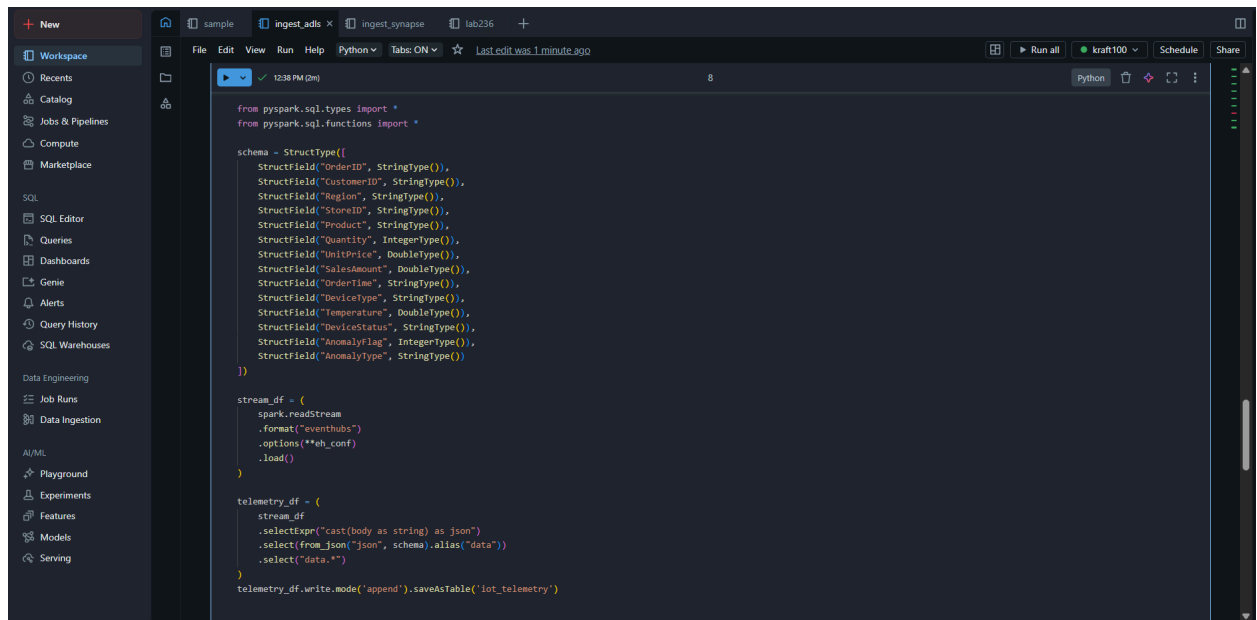
File received on EH



The screenshot shows the Azure Event Hubs Data Explorer interface for the 'sagarhub' instance. The left sidebar contains navigation links: Overview, Access control (IAM), Diagnose and solve problems, Data Explorer (selected), Resource visualizer, Settings, Entities, Features, Automation, and Help. The main area displays 'Total received events: 10' and a 'View next events' link. Below this is a table of events with columns: Sequence Number, Offset, Partition ID, Enqueued Time, Content Type, Message ID, and Event Body. The table shows 10 events, all from Partition ID 0, with timestamps ranging from 12:05:40 PM to 12:05:51 PM. The event bodies are JSON strings representing order data. Below the table, there are tabs for 'Event body' (selected) and 'Event properties'. The 'Event body' tab shows a scrollable list of the event bodies.

Sequence Nu...	Offset	Partition ID	Enqueued Time	Content Type	Message ID	Event Body
0	0	0	Mon, Aug 25, 25, 12:05:40 PM G...			{"OrderID": "a1f3d6b7-45e8"
1	424	0	Mon, Aug 25, 25, 12:05:42 PM G...			{"OrderID": "b7d4f3c2-9c23"
2	840	0	Mon, Aug 25, 25, 12:05:43 PM G...			{"OrderID": "c6e8f5a1-58a9"
3	1264	0	Mon, Aug 25, 25, 12:05:44 PM G...			{"OrderID": "d8f4c9b2-92d2"
4	1688	0	Mon, Aug 25, 25, 12:05:45 PM G...			{"OrderID": "e3c9a2f1-73a4"
5	2112	0	Mon, Aug 25, 25, 12:05:46 PM G...			{"OrderID": "f5d6b8c3-84b9"
6	2552	0	Mon, Aug 25, 25, 12:05:48 PM G...			{"OrderID": "g7a2d1e4-91c5"
7	2968	0	Mon, Aug 25, 25, 12:05:49 PM G...			{"OrderID": "h9f8e3d5-11a2"
8	3384	0	Mon, Aug 25, 25, 12:05:50 PM G...			{"OrderID": "i2d4b7a6-5c31"
9	3800	0	Mon, Aug 25, 25, 12:05:51 PM G...			{"OrderID": "j3e9c5b8-7444"

Reading data from eventhub and storing into my bronze schema as `iot_telemetry`



The screenshot shows a Databricks workspace with a Python script in the editor. The script defines a schema for 'iot_telemetry' and reads data from an Event Hubs instance. The schema includes fields for OrderID, CustomerID, Region, StoreID, Product, Quantity, UnitPrice, SalesAmount, OrderTime, DeviceType, Temperature, DeviceStatus, AnomalyFlag, and AnomalyType. The script uses 'spark.readStream' to read data from the Event Hubs instance and 'write.mode('append').saveAsTable('iot_telemetry')' to write the data to the bronze schema.

```
from pyspark.sql.types import *
from pyspark.sql.functions import *

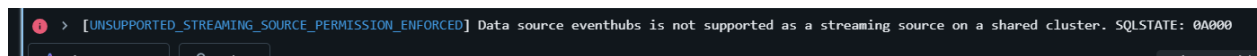
schema = StructType([
    StructField("OrderID", StringType()),
    StructField("CustomerID", StringType()),
    StructField("Region", StringType()),
    StructField("StoreID", StringType()),
    StructField("Product", StringType()),
    StructField("Quantity", IntegerType()),
    StructField("UnitPrice", DoubleType()),
    StructField("SalesAmount", DoubleType()),
    StructField("OrderTime", StringType()),
    StructField("DeviceType", StringType()),
    StructField("Temperature", DoubleType()),
    StructField("DeviceStatus", StringType()),
    StructField("AnomalyFlag", IntegerType()),
    StructField("AnomalyType", StringType())
])

stream_df = (
    spark.readStream
        .format("eventhubs")
        .options(**eh_conf)
        .load()
)

telemetry_df = (
    stream_df
        .selectExpr("cast(body as string) as json")
        .select(from_json("json", schema).alias("data"))
        .select("data.*")
)

telemetry_df.write.mode('append').saveAsTable('iot_telemetry')
```

error i got->



The screenshot shows a Databricks error message: `[UNSUPPORTED_STREAMING_SOURCE_PERMISSION_ENFORCED] Data source eventhubs is not supported as a streaming source on a shared cluster. SQLSTATE: 0A000`.

i solved by making personal cluster .

Data streamed in from event_hub->

The screenshot shows the Databricks interface. On the left is the workspace sidebar with various tools like Workspace, Catalog, Jobs & Pipelines, etc. The main area displays a notebook with a PySpark script that reads from an event hub and writes to a Delta table. Below the script, a table view shows the data loaded into the 'iot_telemetry' table.

OrderID	CustomerID	Region	StoreID	Product	Quantity	UnitPrice
7794cd70-6019-4be0-bce4-5162eae52101	CUST-8517	West	Store-009	Mobile	2	1502.73
6d85d67c-bc5e-43bf-9522-50e645fa5ff8	CUST-7543	North	Store-008	Mobile	5	1012.36
86c2b3ff-076e-4c24-a836-318d69388e0d	CUST-7916	East	Store-013	Tablet	5	1902.44
18179f8f-4017-413a-9c20-c876837ee103	CUST-1188	East	Store-018	Printer	3	1548.77
54a45330-e78f-4d3d-9426-8eb38292bbd3	CUST-1053	North	Store-017	Printer	2	1039.98
f40b3d96-1113-4b72-9a70-3e1dca06544	CUST-4258	West	Store-012	Printer	2	1101.79
f8d9eb0c-6447-4b70-8e46-cb6a04d94e4	CUST-2832	North	Store-010	Mobile	1	519.7
1131c599-e455-4c4d-91c2-d5754ab7ad1d	CUST-2133	West	Store-005	Smartwatch	4	1896.35
307e7b0c-128e-46e0-ad00-89fa1ff2ee83	CUST-4470	West	Store-010	Tablet	3	904.3
a4ef0802-f0d2-40cc-8b05-07d3f3259e94	CUST-6539	East	Store-019	Camera	2	1197.46
ccc9fab-748e-43f5-ba0d-ec19386d4d40	CUST-6413	East	Store-017	Mobile	3	1354.51
a19e0f19-3461-45de-b5e7-f706152d2afc	CUST-8744	West	Store-016	Printer	4	421.3
6669123b-765f-420a-90c2-651e1c3f9f0	CUST-8681	East	Store-004	Laptop	4	1470.82
fc349f6-2650-4e07-b1eb-a4d7d8b1922	CUST-3296	South	Store-006	Headphones	4	537.12

Both table from adls and eventhub stored in my bronze schema of catalog.

iii) Azure synapse ingestion ->

```

# Synapse connection details
synapse_workspace = "sagars"
synapse_pool = "dbi" # your dedicated SQL pool name
synapse_user = "sqladmin"
synapse_password = ""
synapse_jdbc_url = f"jdbc:sqlserver://{synapse_workspace}.sql.azuresynapse.net:1433;database={synapse_pool};"

# Temporary staging location on ADLS Gen2
# This MUST be a storage account accessible by both Synapse and Databricks
temp_storage_account = "sagarstorage08"
temp_container = "syn"
temp_dir = f"abfss://{temp_container}@{temp_storage_account}.dfs.core.windows.net/synapse-dbx-staging"

# Source and Target table names
source_table = "dbo.RetailSales"
target_catalog = "sagar_cat_project1"
target_schema = "bronze"
target_table_name = f"{target_catalog}.{target_schema}.iot_syn"

# 2. Read from Synapse using the connector
print(f"Reading data from Synapse table: {source_table}...")
df = (spark.read
      .format("com.microsoft.sqlserver.jdbc.spark") # Use the Synapse connector
      .option("url", synapse_jdbc_url)
      .option("dbtable", source_table)
      .option("user", synapse_user)
      .option("password", synapse_password)
      .option("tempDir", temp_dir) # Crucial for performance
      .option("forwardSparkAzureStorageCredentials", "true")
      .load())

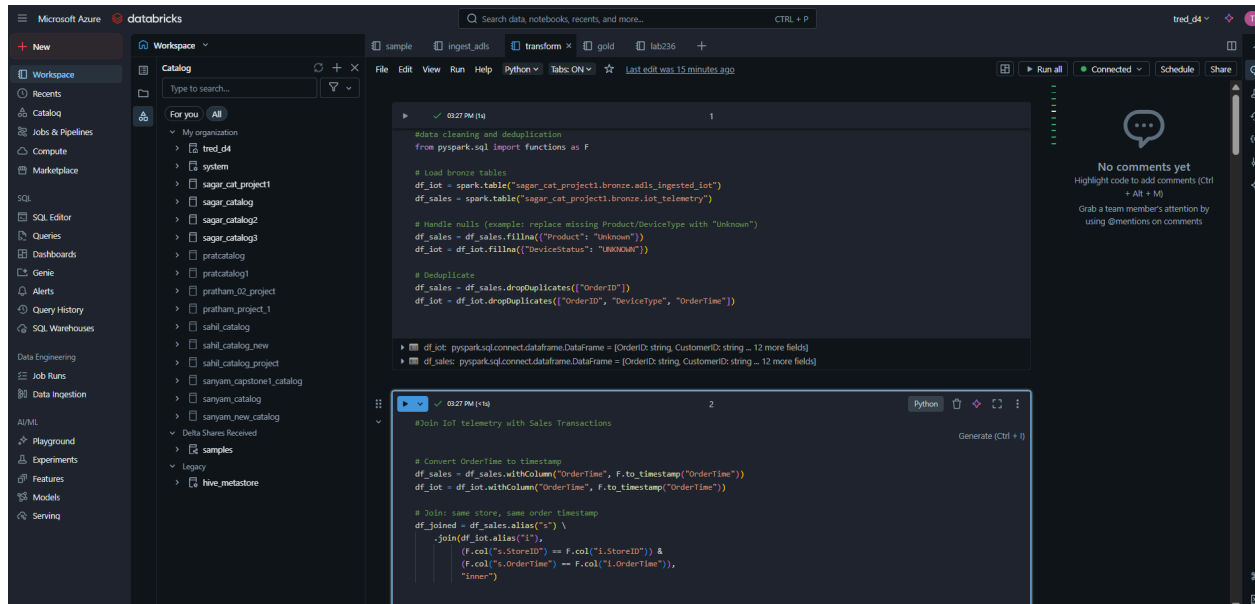
# 3. Write the data as a Delta table in Databricks
print(f"Writing data to Delta table: {target_table_name}...")
(df.write
 .format("delta")
 .mode("overwrite")
 .saveAsTable(target_table_name))

print("Import complete!")

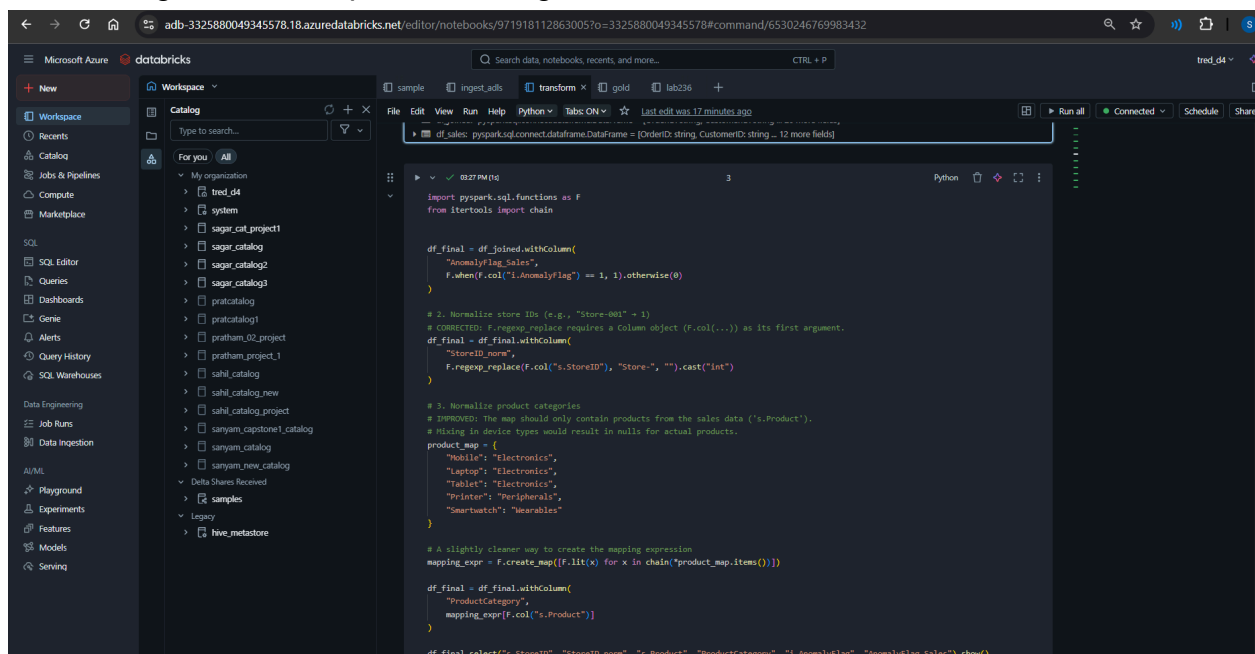
```

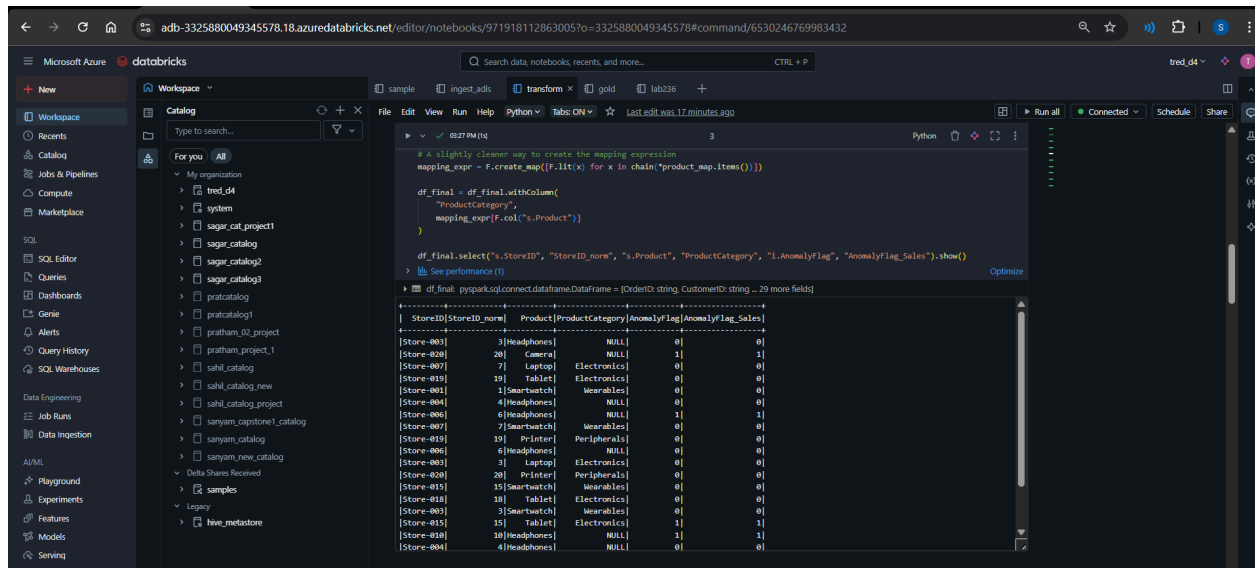
Transformation:

Removal of data quality issues and renaming column



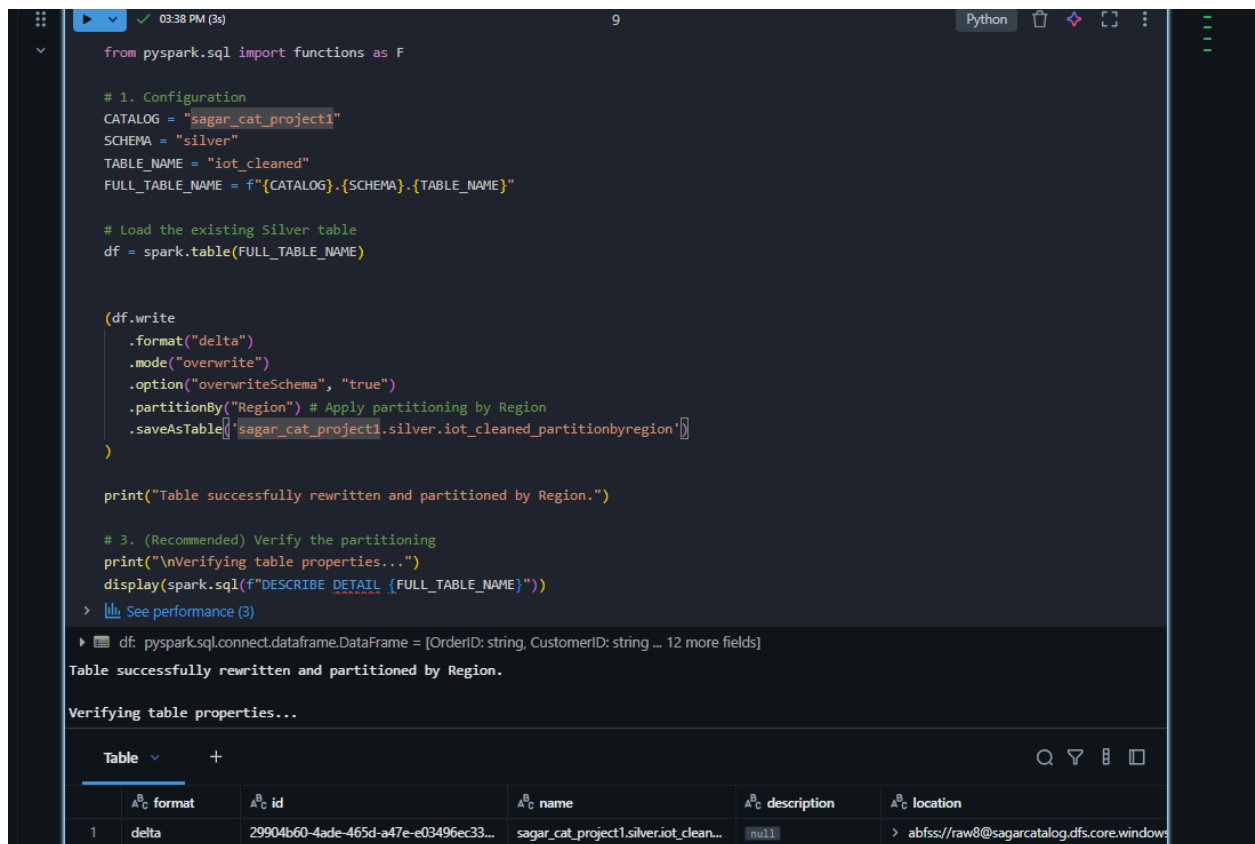
Normalizing storeid and product categories





Finally saving in silver schema of my catalog

Partitioning by region and zordering->



Have a look on all the tables that is created in silver region->

iot_cleaned->

display(spark.table(full_table_name).limit(10))

See performance (3)

--- Displaying top 5 rows of: sagar_cat_project1.silver.iot_cleaned ---

	OrderID	CustomerID	StoreID_norm	ProductCategory	Region	Quantity	UnitPrice	SalesAmount	OrderTime	DeviceType	Temperature	DeviceStatus	AnomalyFlag_Sales	AnomalyType
1	6550279-9c4f-4081-9489-3c5570a0d94	CUST-9476	3	Wearables	East	3	833.22	2499.66	2023-01-08T23:27:07.000+00...	POS_Terminal	16.43	OK	0	
2	91942069-91ca-4099-801d-cd38085c8aab	CUST-2373	20	Wearables	North	9	552.72	4974.48	2023-01-01T06:22:00.000+00...	Temp_Sensor	23.21	MISMATCH	1	Rfid_Mismatch
3	019d090-61aa-4969-aed5-1317b317c7...	CUST-2660	7	Electronics	West	2	1604.24	3208.48	2023-01-16T01:46:21.000+00...	POS_Terminal	21.4	OK	0	
4	762336e-584c-4a3d-8993-2071c10e6d...	CUST-5044	19	Electronics	North	4	952.26	3809.04	2023-01-02T21:52:13.000+00...	Temp_Sensor	16.64	OK	0	
5	28b37595-2556-42a9-8955-c871ca9f48ba	CUST-4842	1	Wearables	East	1	1618.42	1618.42	2023-01-28T20:56:11.000+00...	Temp_Sensor	21.46	OK	0	
6	10dca54-15db-4caf-aef6-58212b3d3574	CUST-2943	4	Wearables	West	1	447.28	447.28	2023-01-14T19:01:10.000+00...	Rfid_Scanner	23.64	OK	0	
7	d44c24b6-5477-4d8c-2847-5833a5d992	CUST-2650	6	Wearables	South	10	758.98	7589.8	2023-01-15T18:36:04.000+00...	Rfid_Scanner	16.68	MISMATCH	1	Rfid_Mismatch
8	7714d928-5a6d-4a2b-809a-a6d79f93a04a	CUST-2609	7	Wearables	West	1	945.77	945.77	2023-01-26T06:34:53.000+00...	Rfid_Scanner	18.03	OK	0	
9	8372d889-946a-46a3-478d-6796464101b	CUST-9466	19	Peripherals	North	3	1125.51	3376.53	2023-01-30T11:45:01.000+00...	Rfid_Scanner	24.38	OK	0	
10	bc9ea752-838f-4153-0a86-768557a33c...	CUST-1472	6	Wearables	West	5	76.67	383.35	2023-01-03T19:20:36.000+00...	Rfid_Scanner	19.57	OK	0	

10 rows | 2.25s runtime

--- Displaying top 5 rows of: sagar_cat_project1.silver.iot_cleaned_partitionbyregion ---

after partitioning->

10 rows | 2.25s runtime

--- Displaying top 5 rows of: sagar_cat_project1.silver.iot_cleaned_partitionbyregion ---

	OrderID	CustomerID	StoreID_norm	ProductCategory	Region	Quantity	UnitPrice	SalesAmount	OrderTime	DeviceType	Temperature	DeviceStatus	AnomalyFlag_Sales	AnomalyType
1	019d090-61aa-4969-aed5-1317b317c7...	CUST-2660	7	Electronics	West	2	1604.24	3208.48	2023-01-16T01:46:21.000+00...	POS_Terminal	21.4	OK	0	
2	10dca54-15db-4caf-aef6-58212b3d3574	CUST-2943	4	Wearables	West	1	447.28	447.28	2023-01-14T19:01:10.000+00...	Rfid_Scanner	23.64	OK	0	
3	7714d928-5a6d-4a2b-809a-a6d79f93a04a	CUST-2609	7	Wearables	West	1	945.77	945.77	2023-01-26T06:34:53.000+00...	Rfid_Scanner	18.03	OK	0	
4	bc9ea752-838f-4153-0a86-768557a33c...	CUST-1472	6	Wearables	West	5	76.67	383.35	2023-01-03T19:20:36.000+00...	Rfid_Scanner	19.57	OK	0	
5	a184d40-14c3-43ea-dc09-ba6800000140	CUST-5436	18	Electronics	West	1	303.36	303.36	2023-01-29T04:11:45.000+00...	Rfid_Scanner	23.99	OK	0	
6	eae95989-a152-418f-9666-ac510772e61	CUST-4259	10	Wearables	West	10	1362.73	13627.3	2023-01-24T15:58:16.000+00...	Temp_Sensor	21.13	MISMATCH	1	Rfid_Mismatch
7	2bee1ee0-d822-4c30-895d-d8e11400000	CUST-4610	11	Electronics	West	3	1057.18	3171.54	2023-01-01T11:19:42.000+00...	Temp_Sensor	23.83	OK	0	
8	307e70c6-128e-49e0-a0d0-899a100ea3	CUST-4470	10	Electronics	West	3	904.3	2712.9	2023-01-26T03:01:10.000+00...	POS_Terminal	18.85	OK	0	
9	451212d0-6a42-49c6-a854-5760676275...	CUST-5056	10	Wearables	West	1	1825.26	1825.26	2023-01-04T04:18:47.000+00...	POS_Terminal	16.8	OK	0	
10	ad76da2-17be-4848-903a-72abf532	CUST-2946	4	Electronics	West	3	1600.05	4800.15	2023-01-17T11:03:46.000+00...	Temp_Sensor	24.57	OK	0	

10 rows | 2.25s runtime

partitioning zorder->

10 rows | 2.25s runtime

--- Displaying top 5 rows of: sagar_cat_project1.silver.iot_cleaned_zorder ---

	OrderID	CustomerID	StoreID	ProductCategory	Region	Quantity	UnitPrice	SalesAmount	OrderTime	DeviceType	Temperature	DeviceStatus	AnomalyFlag_Sales	AnomalyType
1	323c381e-15ec-4f15-bc94-18fdd709fed	CUST-5643	2	Wearables	West	3	652.34	1957.02	2023-01-06T23:50:45.000+00...	POS_Terminal	24.96	OK	0	
2	adb193e5-867e-467e-bbda-097ab3e089...	CUST-4966	6	Electronics	West	5	555.67	2778.35	2023-01-06T03:53:03.000+00...	POS_Terminal	35.33	OVER-HEAT	1	Overheat
3	93205e99-7804-415c-bef0-d74d78e70a3	CUST-3960	1	Wearables	West	4	1611	6444	2023-01-06T23:21:28.000+00...	Rfid_Scanner	18.9	OK	0	
4	4210a853-ded6-4a09-915c-d502837981...	CUST-3598	17	Electronics	West	1	708.58	708.58	2023-01-06T12:47:01.000+00...	Temp_Sensor	19.59	OK	0	
5	b1c9ad00-a05b-4a05-aef6-c958073a02d...	CUST-5748	19	Wearables	West	13	710.43	9235.59	2023-01-06T10:51:25.000+00...	Temp_Sensor	21.35	MISMATCH	1	Rfid_Mismatch
6	e761c16-5ac1-484f-9307-9544372a054	CUST-5776	18	Peripherals	West	4	475.85	1903.4	2023-01-06T03:47:23.000+00...	POS_Terminal	23.01	OK	0	
7	e839eb06-a169-4779-b3b1-ca6c7ceeb3...	CUST-5291	15	Wearables	West	4	1143.77	4575.08	2023-01-06T10:20:33.000+00...	Rfid_Scanner	24.62	OK	0	
8	716ae055-7f0e-4ac3-a136-bc35010c3ed3	CUST-5673	6	Peripherals	West	2	364.45	728.9	2023-01-06T16:13:06.000+00...	Rfid_Scanner	24.96	OK	0	
9	2a2c0a42-318b-49a2-8913-081042c50b...	CUST-1931	20	Wearables	West	1	1430.75	1430.75	2023-01-06T19:45:08.000+00...	Rfid_Scanner	24	OK	0	
10	e65f6a07-c39f-487e-839e-c1c6dc5d90e	CUST-9405	3	Electronics	North	4	220.25	881	2023-01-19T23:10:41.000+00...	Rfid_Scanner	20.02	OK	0	

10 rows | 2.25s runtime

Aggregation & Enrichment Layer (Gold)->

```
from pyspark.sql import functions as F
from pyspark.sql.window import Window

CATALOG = "sagar_cat_project1"
SILVER_SCHEMA = f"{CATALOG}.silver"
GOLD_SCHEMA = f"{CATALOG}.gold"
BASE_TABLE = f"{SILVER_SCHEMA}.iot_cleaned"

# Create the Gold schema if it doesn't exist
spark.sql(f"CREATE SCHEMA IF NOT EXISTS {GOLD_SCHEMA}")
```



```

# Load the base Silver table
base_df = spark.table(BASE_TABLE)

# Standardize column names to ensure consistency
# (Handles cases like 'sales_amount' vs 'SalesAmount' and 'StoreID_norm' vs 'StoreID')
if "sales_amount" in base_df.columns and "SalesAmount" not in base_df.columns:
    base_df = base_df.withColumnRenamed("sales_amount", "SalesAmount")
if "StoreID_norm" in base_df.columns and "StoreID" not in base_df.columns:
    base_df = base_df.withColumnRenamed("StoreID_norm", "StoreID")

# Ensure correct data types for calculations
base_df = (base_df
    .withColumn("OrderTime", F.to_timestamp("OrderTime"))
    .withColumn("SalesAmount", F.col("SalesAmount").cast("double"))
    .withColumn("Quantity", F.col("Quantity").cast("int"))
)

# Derive common date and month grains for aggregation
base_df = (base_df
    .withColumn("order_date", F.to_date("OrderTime"))
    .withColumn("order_month", F.date_trunc("month", "OrderTime"))
)

# -----
# 1) KPI: Daily and Monthly Sales per Region
# -----

print("Creating sales KPIs per region...")

# Daily aggregation
daily_sales = (base_df
    .groupBy("order_date", "Region")
    .agg(
        F.sum("SalesAmount").alias("total_sales"),
        F.sum("Quantity").alias("total_units_sold"),

```

```

        F.countDistinct("OrderID").alias("distinct_orders")
    )
)
(daily_sales.write.format("delta").mode("overwrite")
 .saveAsTable(f"{GOLD_SCHEMA}.daily_sales_by_region"))

# Monthly aggregation
monthly_sales = (base_df
    .groupBy("order_month", "Region")
    .agg(
        F.sum("SalesAmount").alias("total_sales"),
        F.sum("Quantity").alias("total_units_sold"),
        F.countDistinct("OrderID").alias("distinct_orders")
    )
)
(monthly_sales.write.format("delta").mode("overwrite")
 .saveAsTable(f"{GOLD_SCHEMA}.monthly_sales_by_region"))

# -----
# 2) KPI: Device Anomaly Trend per Store
# -----
print("Creating device anomaly trend KPI...")

anomaly_trend = (base_df
    .groupBy("order_date", "StoreID")
    .agg(
        F.sum("AnomalyFlag_Sales").alias("anomaly_count"),
        F.count("OrderID").alias("total_transactions")
    )
    .withColumn("anomaly_rate", F.col("anomaly_count") /
F.col("total_transactions"))
)
(anomaly_trend.write.format("delta").mode("overwrite")
 .saveAsTable(f"{GOLD_SCHEMA}.daily_anomaly_trend_by_store"))

# -----
# 3) KPI: Conversion Rate Impact from Device Failures
# -----

```

```

print("Creating conversion rate impact KPI...")

# We define "conversion" as the ratio of successful orders to total orders at a
store-day level
daily_impact = (base_df
    .groupBy("order_date", "StoreID")
    .agg(
        F.sum(F.when(F.col("AnomalyFlag_Sales") == 1,
1).otherwise(0)).alias("anomaly_orders"),
        F.count("OrderID").alias("total_orders")
    )
    .withColumn("successful_orders", F.col("total_orders") -
F.col("anomaly_orders"))
    .withColumn("success_rate", F.col("successful_orders") /
F.col("total_orders"))
    .withColumn("had_anomaly_day", F.when(F.col("anomaly_orders") > 0,
1).otherwise(0))
)
(daily_impact.write.format("delta").mode("overwrite")
    .saveAsTable(f"{GOLD_SCHEMA}.daily_conversion_impact_by_store"))

# -----
# 4) Enrichment: Tag Stores into Performance Tiers
# -----

print("Enriching stores with performance tiers...")

# Calculate total sales per store for each month
store_monthly_sales = base_df.groupBy("order_month",
"StoreID").agg(F.sum("SalesAmount").alias("total_sales"))

# Use a window function to rank stores within each month
w_spec = Window.partitionBy("order_month").orderBy(F.col("total_sales").desc())
store_ranks = store_monthly_sales.withColumn("rank",
F.percent_rank().over(w_spec))

# Assign tiers based on percentile rank
store_tiers = store_ranks.withColumn("performance_tier",
    F.when(F.col("rank") <= 0.2, "High")

```

```

        .when((F.col("rank") > 0.2) & (F.col("rank") <= 0.8), "Medium")
        .otherwise("Low")
    )
    (store_tiers.select("order_month", "StoreID", "total_sales",
"performance_tier")
        .write.format("delta").mode("overwrite")
        .saveAsTable(f"{GOLD_SCHEMA}.monthly_store_performance_tiers"))

# -----
# 5) Enrichment: Compute Weighted Sales vs. Anomalies Score
# -----
print("Enriching stores with a weighted performance score...")

# Get monthly sales and anomaly counts per store
monthly_kpis = base_df.groupBy("order_month", "StoreID").agg(
    F.sum("SalesAmount").alias("sales"),
    F.sum("AnomalyFlag_Sales").alias("anomalies")
)

# Define a window to normalize scores across all stores within a month
w_monthly = Window.partitionBy("order_month")

# Calculate min-max normalized scores for sales and anomalies
normalized_kpis = (monthly_kpis
    .withColumn("max_sales", F.max("sales").over(w_monthly))
    .withColumn("min_sales", F.min("sales").over(w_monthly))
    .withColumn("max_anomalies", F.max("anomalies").over(w_monthly))
    .withColumn("min_anomalies", F.min("anomalies").over(w_monthly))
    .withColumn("sales_norm",
        F.when(F.col("max_sales") == F.col("min_sales"), 1.0)
        .otherwise((F.col("sales") - F.col("min_sales")) / (F.col("max_sales")
- F.col("min_sales"))))
    )
    .withColumn("anomalies_norm",
        F.when(F.col("max_anomalies") == F.col("min_anomalies"), 0.0)
        .otherwise((F.col("anomalies") - F.col("min_anomalies")) /
(F.col("max_anomalies") - F.col("min_anomalies"))))
    )

```

```
)

# Compute the final weighted score (e.g., 80% weight for sales, 20% for
anomalies)
weighted_score = normalized_kpis.withColumn(
    "performance_score",
    (0.8 * F.col("sales_norm")) - (0.2 * F.col("anomalies_norm"))
)

(weighted_score.select("order_month", "StoreID", "sales", "anomalies",
"performance_score")
.write.format("delta").mode("overwrite")
.saveAsTable(f"{GOLD_SCHEMA}.monthly_store_weighted_score"))

print("\n--- Gold Layer Generation Complete ---")
```

Finally have a look of all the tables that is created.

The screenshot shows the Databricks workspace interface. On the left, the 'Catalog' pane lists various tables and schemas. The main area displays the execution of a Python script. The script output shows the creation of three tables:

- sagar_cat_project1.gold.daily_sales_by_region**: A table with 5 rows and 5 columns: order_date, Region, total_sales, total_units_sold, and distinct_orders.
- sagar_cat_project1.gold.monthly_sales_by_region**: A table with 4 rows and 5 columns: order_month, Region, total_sales, total_units_sold, and distinct_orders.
- sagar_cat_project1.gold.daily_anomaly_trend_by_store**: A table with 4 rows and 5 columns: order_month, Region, total_sales, total_units_sold, and distinct_orders.

Microsoft Azure databricks

Workspace

Catalog

For you (All)

- My organization
 - tred_d4
 - system
 - sagar_cat_project1
 - sagar_catalog
 - sagar_catalog2
 - sagar_catalog3
 - praticatalog
 - praticatalog1
 - pratham_02_project
 - pratham_project_1
 - sahil_catalog
 - sahil_catalog_new
 - sanyam_capstone1_catalog
 - sanyam_catalog
 - sanyam_new_catalog
 - Delta Shares Received
 - samples
 - Legacy
 - hive_metastore

File Edit View Run Help Python Tabs ON ☆ Last edit was 1 minute ago

4 2023-01-01T00:00:00.000-00:00 West 769714779999999 787 234

4 rows | 5.16s runtime Refreshed 45 minutes ago

--- Displaying contents of: sagar_cat_project1.gold.daily_anomaly_trend_by_store ---

| order_date | StoreID | anomaly_count | total_transactions | anomaly_rate |
|------------|---------|---------------|--------------------|--------------|
| 2023-01-29 | 18 | 0 | 1 | 0 |
| 2023-01-05 | 15 | 0 | 1 | 0 |
| 2023-01-26 | 10 | 0 | 3 | 0 |
| 2023-01-20 | 16 | 0 | 2 | 0 |
| 2023-01-11 | 17 | 0 | 2 | 0 |

5 rows | 5.16s runtime Refreshed 45 minutes ago

--- Displaying contents of: sagar_cat_project1.gold.daily_conversion_impact_by_store ---

| order_date | StoreID | anomaly_orders | total_orders | successful_orders | success_rate | had_anomaly_day |
|------------|---------|----------------|--------------|-------------------|--------------|-----------------|
| 2023-01-29 | 18 | 0 | 1 | 1 | 1 | 0 |
| 2023-01-05 | 15 | 0 | 1 | 1 | 1 | 0 |
| 2023-01-26 | 10 | 0 | 3 | 3 | 1 | 0 |
| 2023-01-20 | 16 | 0 | 2 | 2 | 1 | 0 |
| 2023-01-11 | 17 | 0 | 2 | 2 | 1 | 0 |

5 rows | 5.16s runtime Refreshed 45 minutes ago

--- Displaying contents of: sagar_cat_project1.gold.monthly_store_performance_tiers ---

Microsoft Azure databricks

Workspace

Catalog

For you (All)

- My organization
 - tred_d4
 - system
 - sagar_cat_project1
 - sagar_catalog
 - sagar_catalog2
 - sagar_catalog3
 - praticatalog
 - praticatalog1
 - pratham_02_project
 - pratham_project_1
 - sahil_catalog
 - sahil_catalog_new
 - sanyam_capstone1_catalog
 - sanyam_catalog
 - sanyam_new_catalog
 - Delta Shares Received
 - samples
 - Legacy
 - hive_metastore

File Edit View Run Help Python Tabs ON ☆ Last edit was 1 minute ago

1 2023-01-29 18 0 1 1 1 0

2 2023-01-05 15 0 1 1 1 0

3 2023-01-26 10 0 3 3 1 0

4 2023-01-20 16 0 2 2 1 0

5 2023-01-11 17 0 2 2 1 0

5 rows | 5.16s runtime Refreshed 45 minutes ago

--- Displaying contents of: sagar_cat_project1.gold.monthly_store_performance_tiers ---

| order_month | StoreID | total_sales | performance_tier |
|-------------------------------|---------|--------------------|------------------|
| 2023-01-01T00:00:00.000-00:00 | 15 | 197270.27999999999 | High |
| 2023-01-01T00:00:00.000-00:00 | 17 | 184736.13000000000 | High |
| 2023-01-01T00:00:00.000-00:00 | 10 | 184143.77 | High |
| 2023-01-01T00:00:00.000-00:00 | 13 | 176925.58999999999 | High |
| 2023-01-01T00:00:00.000-00:00 | 4 | 175538.23 | Medium |

5 rows | 5.16s runtime Refreshed 45 minutes ago

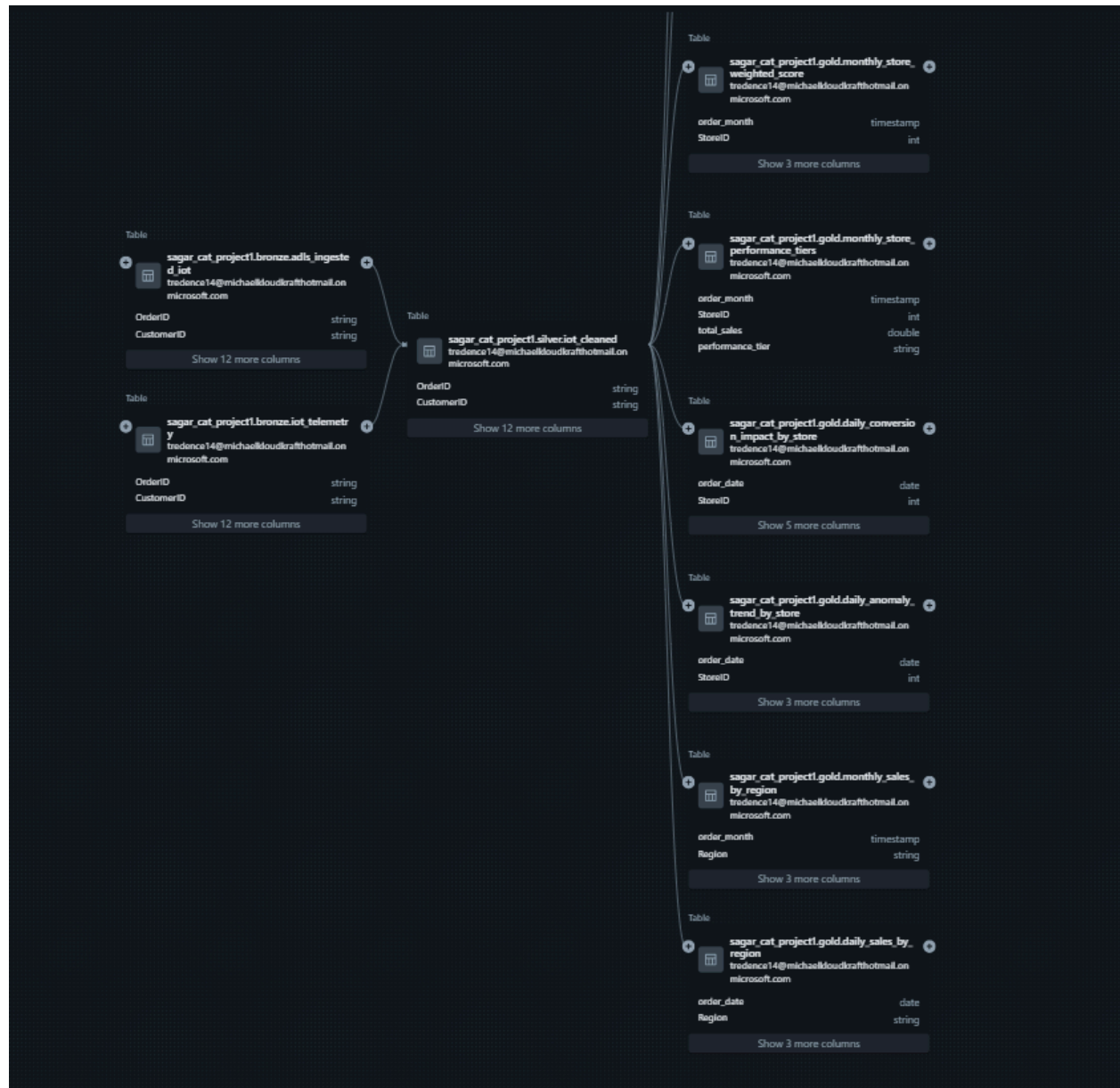
--- Displaying contents of: sagar_cat_project1.gold.monthly_store_weighted_score ---

| order_month | StoreID | total_sales | anomalies | performance_score |
|-------------------------------|---------|--------------------|-----------|--------------------|
| 2023-01-01T00:00:00.000-00:00 | 18 | 173029.89 | 3 | 0.5854040612526225 |
| 2023-01-01T00:00:00.000-00:00 | 11 | 135648.96999999999 | 1 | 0.367662552392394 |
| 2023-01-01T00:00:00.000-00:00 | 5 | 174334.11000000000 | 8 | 0.4835234070595928 |
| 2023-01-01T00:00:00.000-00:00 | 12 | 152687.06 | 2 | 0.4648021774526915 |
| 2023-01-01T00:00:00.000-00:00 | 1 | 163738.09 | 5 | 0.4750479960972901 |

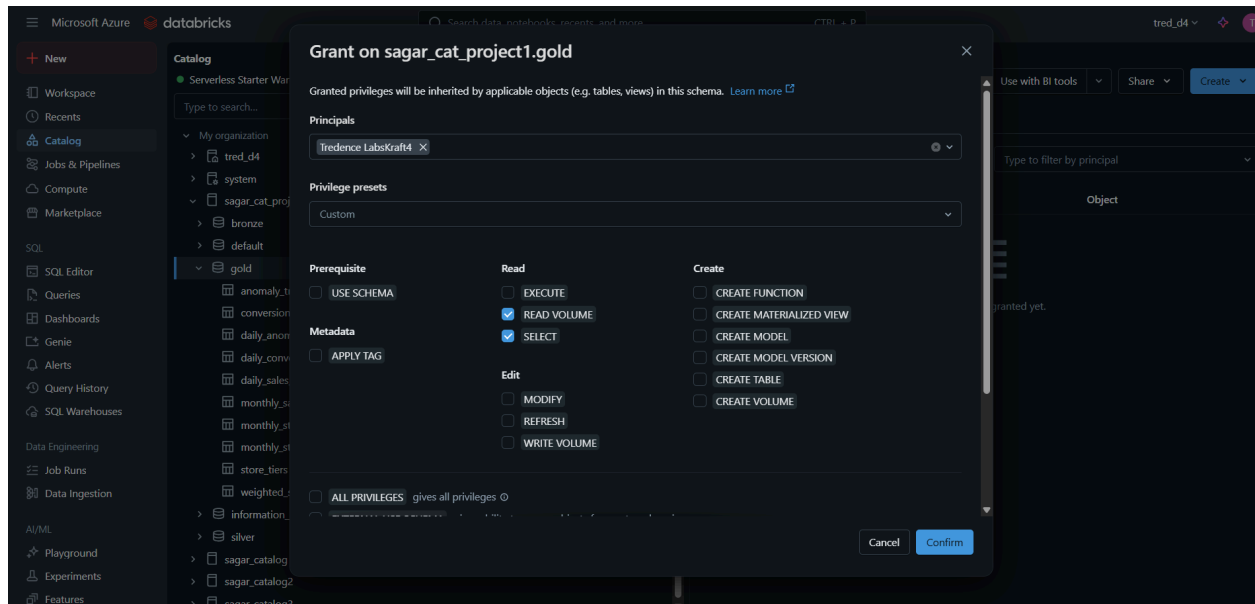
5 rows | 5.16s runtime Refreshed 45 minutes ago

JOBS creation->

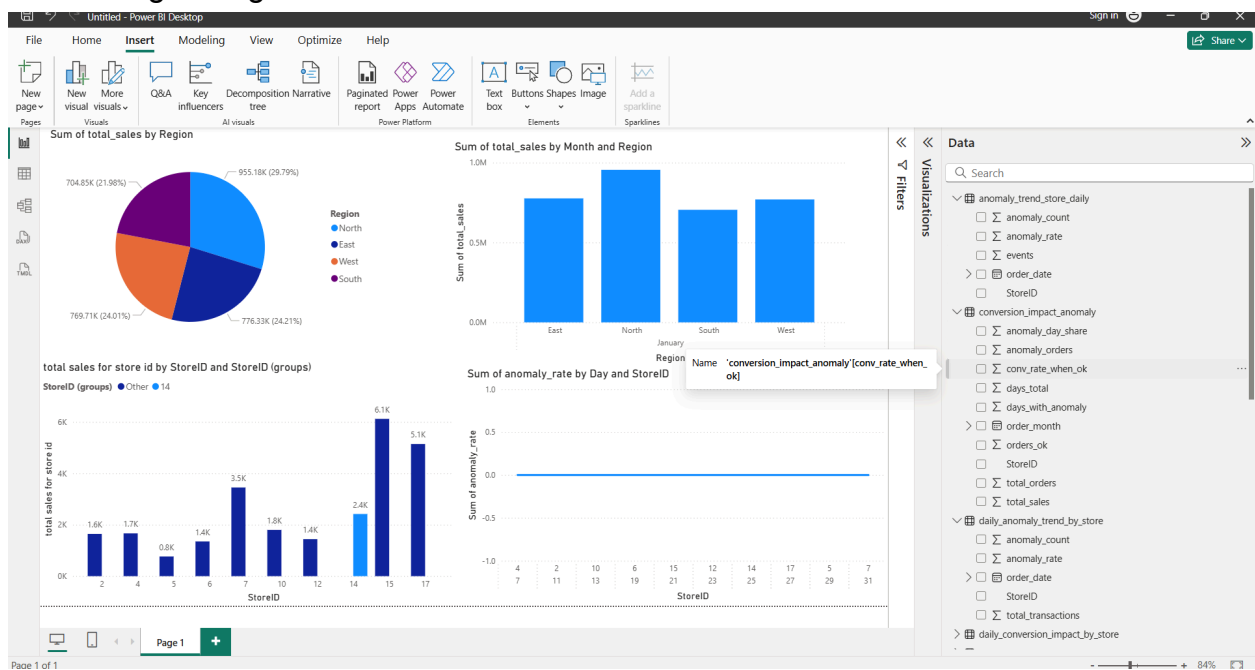
Lineage Graph->

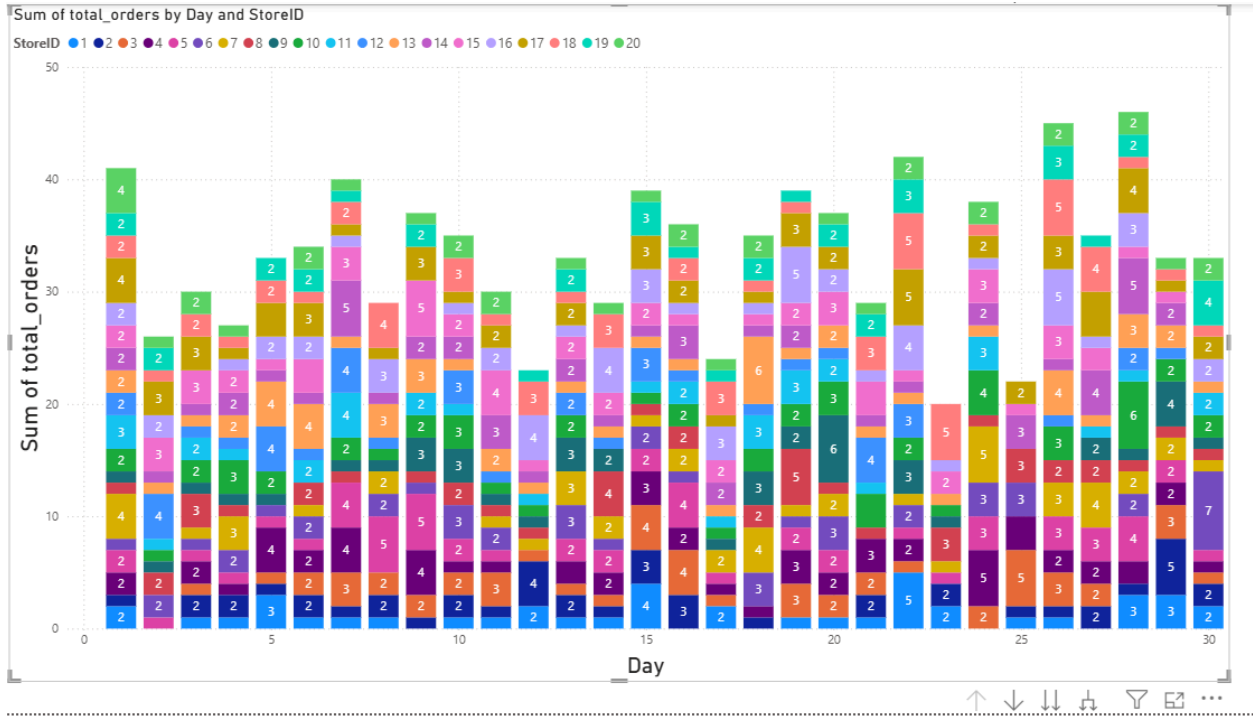


Giving bi user access to my gold tables->



Connecting through PowerBi->





Sum of weighted_score by StoreID

